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# Babar Ali

AI/ML Engineer

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## PROFESSIONAL SUMMARY

AI/ML Engineer specializing in agentic pipelines, LLM fine-tuning (LoRA/QLoRA), and multimodal systems. Experienced in multi-agent orchestration, real-time inference deployment, and production-grade deep learning across medical imaging and NLP domains.

## EXPERIENCE

**AI Engineer** *Neural Lines — 1 year*

Built NLP pipelines for text classification, NER, and sentiment analysis. Engineered data preprocessing workflows and delivered insights through automated visualization and model evaluation reports.

## TECHNICAL SKILLS

**Languages:** Python, C++

**Frameworks & Libraries:** PyTorch, TensorFlow, LangChain, LangGraph, LlamaIndex, Transformers (HuggingFace), Diffusers, OpenCV, Pydantic AI

**LLMOps & Deployment:** vLLM, Ollama, FastAPI, Docker, ONNX Export, Modal, Replicate, Weights & Biases

**AI Concepts:** Agentic AI & Multi-Agent Systems, LLM Fine-Tuning (LoRA/QLoRA), Agentic RAG, Multimodal AI, RL (PPO/DQN), Computer Vision, NLP, Prompt Engineering, Model Quantization

## PROJECTS

**Multi-Agent AI Pipeline for VR Mental Health** *VR System — 2025*

- Architected a clinical-grade VR mental health system using fine-tuned **Mistral AI** as conversational core, orchestrated via **LangGraph**; deployed four parallel **BERT classifiers** for intent detection, emotion recognition, risk assessment, and topic classification
- Integrated an **XAI-augmented sentiment analyzer** and custom **TextCNN** for response quality scoring — entire multi-agent pipeline operates in real-time within the VR rendering loop

**Autonomous Driving Agent via Reinforcement Learning** *CARLA Simulator — 2024*

- Trained a **PPO-based** self-driving agent in CARLA using raw camera input for lane following and adaptive speed regulation across varied road conditions
- Engineered a **custom reward shaping** function to penalize unsafe behavior and accelerate convergence; optimized the observation pipeline to minimize per-step inference latency

**YOLOv8 Medical Imaging Detection & Segmentation Suite** *Custom Datasets — 2023-2024*

- **COVID-19 classification** from chest X-rays using fine-tuned YOLOv8 detection heads
- **RBC/WBC cell detection** in blood smear microscopy images with instance-level localization
- **Breast cancer segmentation** from ultrasound imagery using YOLOv8-seg with pixel-level masks
- **Knee Osteoarthritis Severity Grading from X-ray Imaging** Built a clinically-motivated deep learning system for automated grading of knee osteoarthritis severity from radiographic images, following the Kellgren-Lawrence (KL) grading standard and ResNet-based multi-class classifier

## EDUCATION

**Bachelor of Science in Artificial Intelligence** Bahria University, Islamabad 2022 – 2026

## OTHER SKILLS

Agentic Workflow Design · Rapid Prototyping · Research-to-Production Implementation · API Integration · Technical Problem Solving · Git & Version Control